

Feeding at flowers

There are more than 250,000 different kinds of flowering plant, and more than three-quarters of them produce nectar, an energy-rich liquid that animals feed on. In return, animals pollinate the flowers by spreading grains of pollen from flower to flower, so that seeds can form. Flower-feeders include huge numbers of insects and some specialized birds and mammals. In addition to nectar, some animals eat pollen, or even flowers themselves.

Night flights

As night falls, daytime pollinators disappear, and different animals set off to visit flowers. These Mexican long-tongued bats are feeding at the flowers of a century plant, which grow in clusters on the end of long stalks. Unlike insect-eating bats, those that visit flowers have good eyesight and a keen sense of smell. To a human nose, bat-pollinated flowers often smell sickly sweet, but their aroma can attract bats from more than half a mile (1 km) downwind.

Food on the wing

Beating its wings more than 50 times a second, a hummingbird stays rock-steady as it feeds at a columbine flower. Its beak reaches deep into the bloom, while its tongue darts in and out to lap up nectar. Despite being just 4 in (10 cm) long, the ruby-throated hummingbird migrates from Central America as far north as Canada, fueling up with nectar before starting its flight.

Rapidly beating wings keep hummingbird directly beneath flower

Beak closed when feeding, except for small gap at tip

Part-time visitors

Using its jaws, a leaf beetle scrapes pollen from the male parts, or stamens, of a flower. It eats most of the pollen grains, but some stick to its body, ready to be transferred to the next flower that it visits. Leaf beetles get only part of their food from flowers. They visit flowers where there is room to clamber around and the pollen or nectar is easy to reach. The rest of their diet comes from eating other parts of the plant, such as the leaves and seeds.

Strong, leathery flowers resist damage caused by feeding bats